Rank Matters More Than You Think

Haddad, Nguyen et al. · 2025

Fictional sample paper generated for the ClaimLens demo.

We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. Code will be released after review. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split.

Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced.

All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds.

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We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps.

This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models.

Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced. We sweep the LoRA rank from 8 to 512. We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section.

All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds.

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We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. We use a single 7B base model throughout. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps.

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Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration.

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We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one.

Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. Table 2 lists the code benchmarks we evaluate. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration.

Table 2

We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. We report pass@1 on every code benchmark. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one.

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We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant.

We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

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Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. On code, LoRA matches full fine-tuning only at rank 256 and above. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. Below rank 64, code accuracy drops sharply. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

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